8.1 Energy Transfer and Deposition

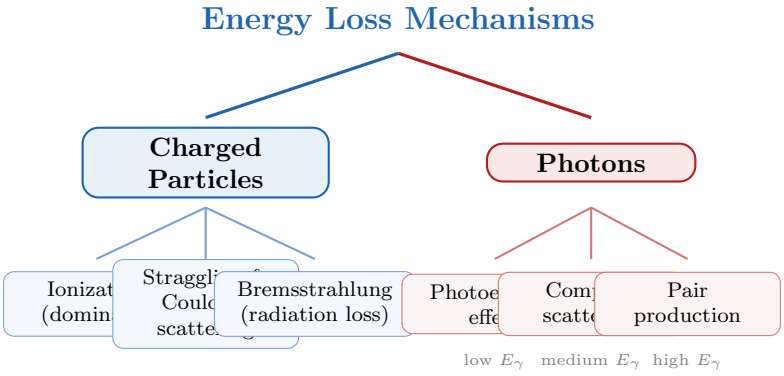
How Particles Lose Energy in Matter—and How We Exploit It

Mann, Chapter 8 · Rewritten with extended explanations and figures

The Big Picture

A detector works by *stealing energy* from the particles passing through it. If a particle didn't transfer energy to the detector material, we would never know it existed. This entire section is about the **mechanisms** by which that energy transfer happens. Everything else in detector physics—calorimeters, tracking chambers, particle ID—is built on top of these mechanisms.

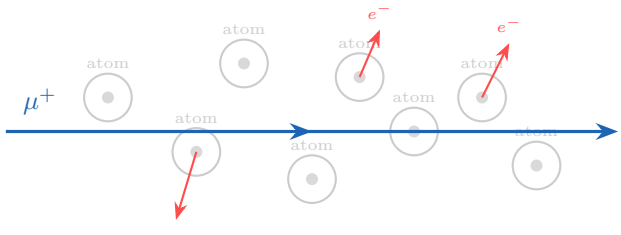
There are two broad categories: **charged particles** (which interact continuously via the electromagnetic force as they traverse the medium) and **photons** (which interact discretely, one event at a time). We treat them separately.



Part I: Charged Particles in Matter

What actually happens microscopically?

Picture a fast charged particle (say a proton or muon) flying through a block of material. It doesn't "hit" anything in the billiard-ball sense. Instead, its electric field *tugs* on the atomic electrons it passes near, transferring small amounts of energy to them. Some of these electrons get knocked free (ionization); others get bumped to excited states.



The particle's primary interactions are with **atomic electrons**, not nuclei. Why? Scattering from nuclei changes the particle's *direction* a lot (they're heavy) but transfers very little *energy* (elastic, like a ping-pong ball bouncing off a bowling ball). Scattering from electrons is *inelastic*—it knocks them free—and transfers real energy.

Direct nuclear collisions

The incoming particle *can* hit a nucleus directly, but the cross section for this is tiny. Nuclei are small targets compared to the atomic electron cloud. We neglect direct nuclear collisions.

The key quantity for everything that follows is the **stopping power**: the rate of energy loss per unit path length.

Stopping Power & Range

$$\text{Stopping power} = -\frac{dE}{dx} \qquad \text{(energy lost per cm traversed)}$$

Once you know dE/dx, you can compute how far the particle travels before it stops—its **range**:

$$R = \int_0^R dx = \int_0^E \left(\frac{dE}{dx}\right)^{-1} dE$$

1. Ionization Loss: The Bethe–Bloch Formula

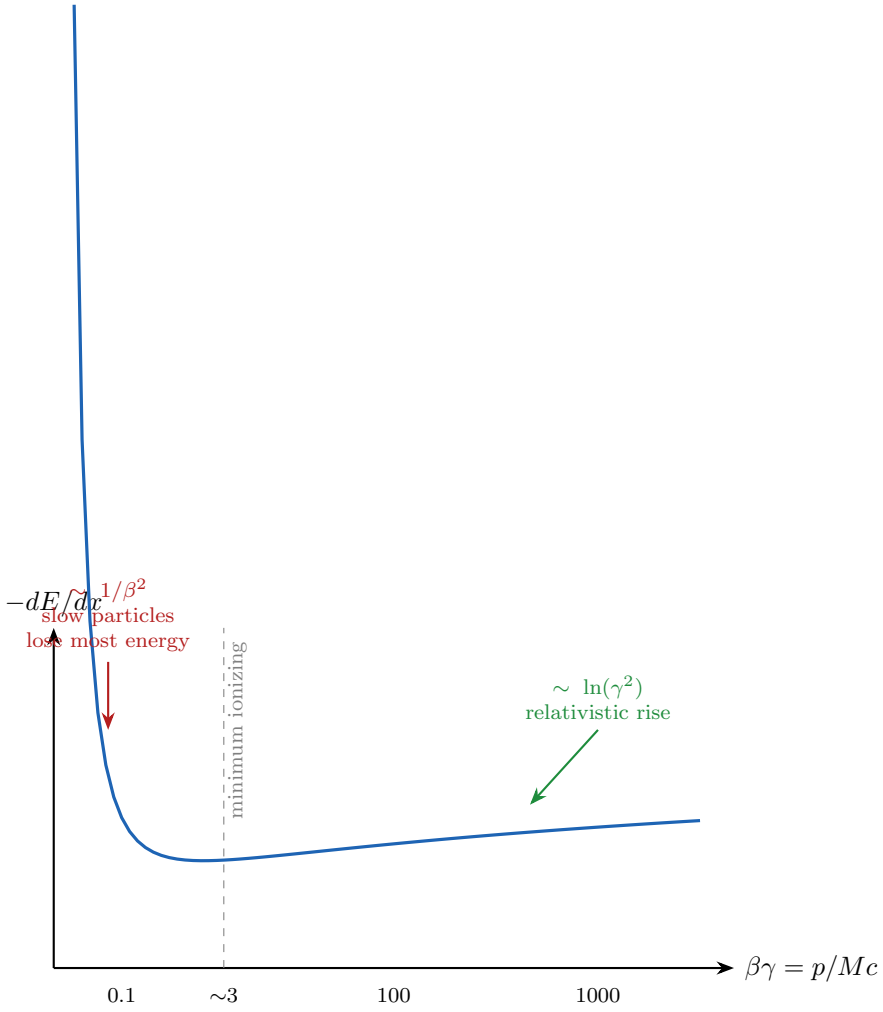
This is the *dominant* energy-loss mechanism for charged particles at most energies. Bethe and Bloch derived a formula from electromagnetism that describes the mean rate of energy loss:

Bethe–Bloch Formula (Mann Eq. 8.2)

$$-\left.\frac{dE}{dx}\right|_{\text{ion}} = \frac{4\pi N_A (ze)^2 e^2}{m_e c^2 \beta^2} \cdot \frac{Z}{A}\left[\ln\left(\frac{2m_e c^2 \beta^2 \gamma^2}{I}\right) - \beta^2\right]$$

- z = charge of incident particle (in units of e)
- $\beta = v/c, \quad \gamma = (1-\beta^2)^{-1/2}$ for the incident particle
- Z, A = atomic number, mass number of the medium
- I = mean ionization potential of the medium ($\sim 10Z$ eV)
- m_e = electron mass, $\quad N_A$ = Avogadro's number

You don't need to derive this—but you *do* need to understand its behavior.



Three regimes of Bethe–Bloch

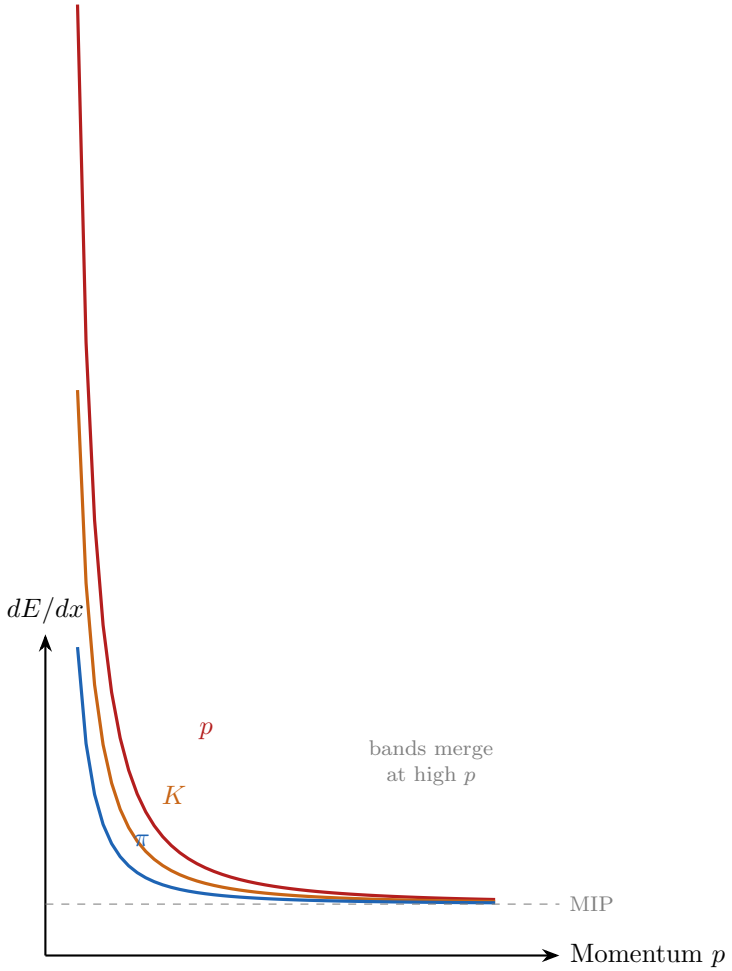
Low β (slow particles): $dE/dx \propto 1/\beta^2$. Slow particles spend more time near each atom, so they lose energy faster. This is why particles deposit the most energy right before they stop (the **Bragg peak**—used in proton therapy for cancer!).
Minimum ionizing ($\beta\gamma \approx 3$): The curve bottoms out. For *all* solid materials, the minimum value is approximately $-dE/dx \approx 2\rho\,\mathrm{MeV/g/cm^2}$. Particles near this minimum are called **minimum ionizing particles (MIPs)**.
High $\beta\gamma$ (relativistic rise): dE/dx grows as $\ln(\gamma^2)$—slowly. Eventually screening effects from nearby atoms cause it to plateau. Above the minimum, you can't easily distinguish particle types by ionization alone.

Why dE/dx can identify particles

At a *given momentum* p, different-mass particles have different β:

$$\beta = \frac{p}{\sqrt{p^2 + M^2c^2}} \qquad \Longrightarrow \qquad \frac{dE}{dx} \propto \frac{1}{\beta^2} \approx \frac{M^2}{p^2} \quad \text{(low } p\text{)}$$

So heavier particles at the same momentum are slower and lose energy faster. Measuring dE/dx vs. p gives you bands that separate π, K, p, etc.



2. Straggling and Coulomb Scattering

The Bethe–Bloch formula gives the *mean* energy loss. But energy transfer is a statistical process: sometimes a lot is transferred, sometimes a little. This fluctuation is called **straggling**.

Mann works through a toy example (Eqs. 8.6–8.13) where the cross section for momentum transfer to the detector medium is

$$\frac{d\sigma}{dq^2} = e^{-8q^2/\lambda^2}$$

with $\lambda \sim 1/(1.2\,A^{1/3})$. The result: the mean kinetic energy transferred to a detector nucleus is $\langle T \rangle = \lambda^2/(16M)$, and the dispersion ΔT *equals* the mean. The punchline: straggling is small for heavy targets (lead: $\Delta T \sim 7\,\text{keV}$) but significant for light ones (hydrogen: $\Delta T \sim 50\,\text{MeV}$).

Physical picture of straggling

Think of it this way: the particle takes many small "sips" of energy from the medium. Each sip is random. The total is well-defined on average (Bethe–Bloch), but any individual particle's total will fluctuate. Heavier target nuclei give smaller, more predictable sips.

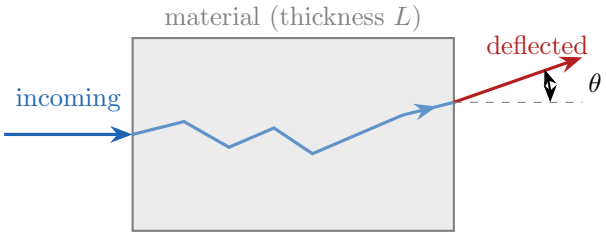
Coulomb scattering is a separate effect: the incident particle's *direction* gets randomized by many small deflections off nuclei. This is called **multiple Coulomb scattering**. The Rutherford formula (Mann Eq. 8.14) tells us $\sigma(\theta)$ is large for small θ—lots of tiny deflections.

The net result after traversing thickness L of material is a Gaussian-distributed scattering angle with r.m.s.:

Multiple Coulomb Scattering (Mann Eq. 8.15)

$$\langle\theta\rangle_{\text{rms}} \approx \frac{20\,\text{MeV} \cdot q}{\beta pc}\sqrt{\frac{L}{X_0}}$$

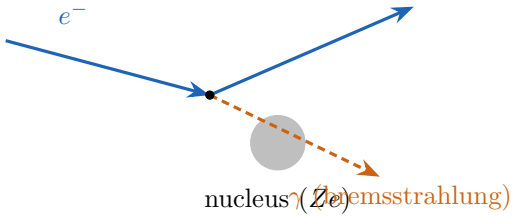
where q is the particle's charge, βpc its momentum, L the material thickness, and X_0 the radiation length.



This matters because it limits how precisely you can measure a particle's direction. The thicker the detector, the worse the angular resolution.

3. Radiation Loss (Bremsstrahlung)

A charged particle that *accelerates* radiates photons. When a particle decelerates in the Coulomb field of a nucleus, this radiation is called **bremsstrahlung** (German: "braking radiation").



The key result: the energy loss per unit length due to bremsstrahlung is *proportional to the particle's energy*:

Bremsstrahlung Energy Loss (Mann Eq. 8.16)

$$-\left.\frac{dE}{dx}\right|_{\text{brem}} = \frac{E}{X_0} \qquad \Longrightarrow \qquad E(x) = E_0\, e^{-x/X_0}$$

The energy decays **exponentially** with distance, with characteristic length X_0 (the **radiation length** of the material).

This is qualitatively different from ionization loss: ionization loss is roughly constant (for MIPs), but radiation loss *grows with energy*. At some point, radiation dominates.

Why electrons and not protons

Bremsstrahlung power scales as $\sim (E/M)^4$ at fixed energy (recall synchrotron radiation from Ch. 7). For a given energy, an electron radiates $(m_p/m_e)^4 \approx 10^{13}$ times more than a proton. This is why radiation loss matters for electrons/positrons and is negligible for heavier particles at accelerator energies.

Ionization vs. Bremsstrahlung: The Critical Energy

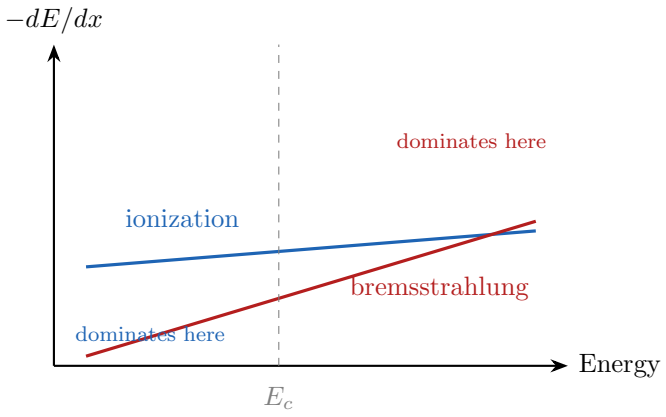
At low energy, ionization dominates. At high energy, bremsstrahlung dominates. The crossover is the **critical energy** E_c.

Critical Energy (Mann Eqs. 8.18–8.19)

$$\frac{(dE/dx)_{\text{brem}}}{(dE/dx)_{\text{ion}}} = \frac{ZE}{1200\, Mc^2}$$

Setting this ratio to 1 and using $M = m_e$:

$$\boxed{E_c \approx \frac{600}{Z}\ \text{MeV} \qquad \text{(for electrons)}}$$



Radiation Length: A Universal Yardstick

The radiation length X_0 characterizes how quickly a material chews up electron/photon energy. It appears in bremsstrahlung, pair production, and shower development.

Material	Z	E_c (MeV)	X_0 (cm)
Liquid H_2	1	340	887
Carbon	6	103	28
Aluminum	13	47	9.0
Iron	26	24	1.77
Lead	82	6.9	0.56

Pattern

High-Z materials have **short** X_0 and **low** E_c. Lead stops electrons fast. Hydrogen barely slows them. This is why calorimeters are built from lead, tungsten, or iron.

Part II: Photons in Matter

Photons are neutral, so they don't continuously ionize the medium. Instead, they interact *discretely*: a single photon is either absorbed, scattered, or converted—all-or-nothing. The beam intensity falls exponentially:

Photon Absorption (Mann Eq. 8.20)

$$I(x) = I_0\, e^{-\mu x}$$

where μ is the **absorption coefficient**. Its inverse μ^{-1} is the mean free path—the average distance a photon travels before something happens to it.

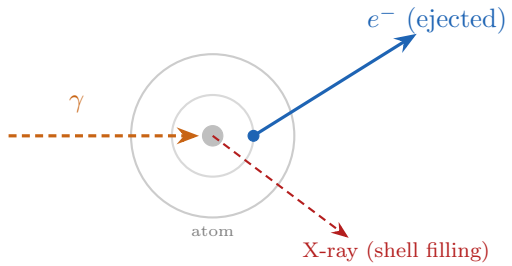
The total absorption coefficient is the sum of contributions from three processes:

$$\mu = \mu_{\text{photo}} + \mu_{\text{Compton}} + \mu_{\text{pair}}$$

Each dominates in a different energy regime.

1. Photoelectric Effect (Low Energy)

A photon is absorbed by a bound atomic electron, which is then ejected. The photon disappears entirely—all its energy goes to the electron's kinetic energy plus the binding energy.



Cross section (Mann Eq. 8.22):

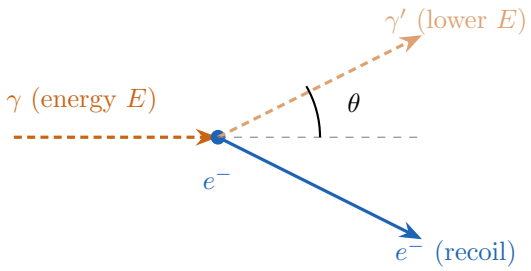
$$\sigma_{\text{photo}} \propto \begin{cases} Z^5/E^{7/2} & E < m_e c^2 \\ Z^5/E & E > m_e c^2 \end{cases}$$

Two things to notice

(1) The Z^5 dependence means lead is $\sim (82/13)^5 \approx 10^4$ times more effective than aluminum at absorbing low-energy photons.
(2) At higher energies the cross section falls, and Compton takes over.

2. Compton Scattering (Intermediate Energy)

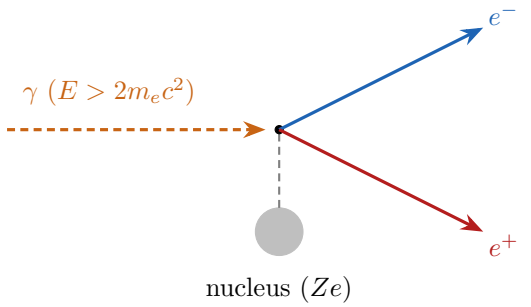
The photon scatters off a quasi-free electron, transferring some of its energy and continuing at a lower frequency. You already know the kinematics from Griffiths Ch. 7.



The cross section is given by the Klein–Nishina formula (the QED calculation you did in Griffiths 7.30/7.41). At low energies it reduces to the classical Thomson cross section; at high energies it falls as $\sim 1/E$.

3. Pair Production (High Energy)

A high-energy photon, in the Coulomb field of a nucleus, converts into an e^+e^- pair. This requires $E_\gamma \geq 2m_ec^2 = 1.022\,\text{MeV}$ (threshold).



Why the nucleus must be there

A photon cannot produce a pair *in vacuum*—it would violate conservation of 4-momentum (you can't satisfy both E and $\vec{p}$ conservation simultaneously for $\gamma \to e^+e^-$ with no recoil partner). The nucleus absorbs the recoil momentum. This is the same reason bremsstrahlung needs a nucleus.

At high energies, pair production dominates, and its cross section is related to the radiation length:

$$\sigma_{\text{pair}} \approx \frac{7}{9} \cdot \frac{1}{nX_0}$$

where n is the number density of nuclei. So the mean free path for pair production is $\frac{9}{7}X_0$—almost the same length scale as bremsstrahlung for electrons. This is not a coincidence: the Feynman diagrams are related by crossing symmetry.

Bremsstrahlung ↔ Pair Production: Crossing Symmetry

These two processes are "crossed" versions of the same QED vertex:

$$\text{Bremsstrahlung:} \quad e^- + Z \to e^- + Z + \gamma$$

$$\text{Pair production:} \quad \gamma + Z \to e^+ + e^- + Z$$

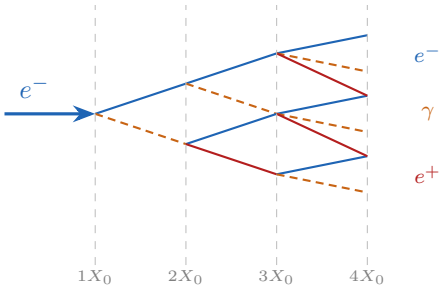
Move the outgoing γ to the initial state and the incoming e^- to a final-state e^+—same matrix element. That's why they share the same length scale X_0.

Energy Dependence: Which Process Wins?



Part III: The Punchline—Electromagnetic Showers

At high energies, bremsstrahlung and pair production *feed each other*:



The number of particles **doubles** every radiation length. After x radiation lengths, the average particle energy is $E/2^x$. The shower grows until the average energy drops to E_c, at which point ionization takes over and the particles stop.

Shower Maximum

$$\langle E \rangle \approx \frac{E_0}{2^x} \qquad \Longrightarrow \qquad x_{\max} = \frac{\ln(E_0/E_c)}{\ln 2} \text{ radiation lengths}$$

A 100 GeV electron in lead ($E_c \approx 7\,\text{MeV}$): $x_{\max} \approx 14\,X_0 \approx 8\,\text{cm}$.

Why this matters for detectors

This is the operating principle of an **electromagnetic calorimeter**. You put a slab of high-Z material (lead, tungsten) in front of light-collecting sensors. The incoming e^- or γ creates a shower; you measure the total light output, which is proportional to the original particle's energy. The shower is compact (a few cm in lead), so the detector doesn't need to be enormous.

Summary: Everything on One Page

Process	Physics	Dominates when	Key formula
Ionization	EM interaction with atomic e^-	Always (charged particles)	Bethe–Bloch
Bremsstrahlung	Radiation in nuclear Coulomb field	$E > E_c$ (electrons)	$E = E_0 e^{-x/X_0}$
Photoelectric	γ absorbed by bound e^-	$E_\gamma \lesssim$ keV	$\sigma \propto Z^5/E^{7/2}$
Compton	γ scatters off e^-	$E_\gamma \sim$ MeV	Klein–Nishina
Pair production	$\gamma \to e^+e^-$ near nucleus	$E_\gamma \gtrsim 10\,\text{MeV}$	$\lambda \approx \frac{9}{7} X_0$

Key Numbers to Remember

MIP stopping power: $-dE/dx \approx 2\rho$ MeV/g/cm^2

Critical energy (electrons): $E_c \approx 600/Z$ MeV

Radiation length (Pb): $X_0 = 0.56$ cm

Shower max: $x_{\max} = \ln(E/E_c)/\ln 2$ radiation lengths

The Detector Design Logic

Every layer of a modern detector exploits a specific mechanism from this section:

- **Tracking chamber** (inner): thin material, measures ionization → reconstructs trajectories and momenta.
- **EM calorimeter**: thick high-Z material, uses bremsstrahlung + pair production → measures e^- and γ energies via showers.
- **Hadron calorimeter**: even thicker, uses nuclear interactions (analogous cascade) → measures hadron energies.
- **Muon chambers** (outer): muons are too heavy for bremsstrahlung and don't interact strongly, so they punch through everything else. Detecting them last is the identification.